
Exam formula list: must-know relations

Planetary Systems final exam • closed book

Planetary Systems (WBAS002-05) • Kapteyn Astronomical Institute, University of Groningen

The final exam is closed book: a calculator and a pen are all that is allowed. The twelve relations on this list are the ones you are expected to know from memory and apply without prompting. Every other formula a question needs, for example the vis-viva equation, the Roche limit, or the Jeans escape flux, is printed in the question itself. Each entry below states where the relation is introduced in the [lecture notes](#); the same list with the same links appears on the [mock exams page](#).

1. Newton's law of gravitation

$$F = \frac{GMm}{r^2}, \quad U = -\frac{GMm}{r}$$

The attractive force between masses M and m at separation r , and the corresponding gravitational potential energy. This pair underlies all orbital dynamics and the energy release of accretion and differentiation. The force law drives the orbit derivations of [Lecture 1](#); the energy form is introduced in [Lecture 2](#) and drives the heating budgets of [Lecture 4](#).

2. Circular orbit speed

$$v_c = \sqrt{\frac{GM}{r}}$$

The speed of a body on a circular orbit of radius r around a central mass M , from the balance of gravitational and centripetal acceleration. Derived in [Lecture 1](#); generalised by the vis-viva equation of [Lecture 2](#).

3. Escape velocity

$$v_{\text{esc}} = \sqrt{\frac{2GM}{r}}$$

The minimum speed needed to escape to infinity from distance r , from setting kinetic plus potential energy to zero. It controls impact energetics and which gases a planet can retain. Introduced in [Lecture 2](#); central to atmospheric escape in [Lecture 5](#).

4. Kepler's third law

$$p^2 = \frac{4\pi^2 a^3}{GM}$$

The orbital period P of an orbit with semi-major axis a around a central mass M . It is the primary tool for weighing central bodies and for converting exoplanet periods into orbital distances. Derived in Newtonian form in [Lecture 1](#); restated in [Lecture 2](#) and applied to exoplanets in [Lecture 13](#).

5. Mean density

$$M = \frac{4}{3}\pi R^3 \bar{\rho}$$

The relation between mass, radius, and bulk density $\bar{\rho}$. Bulk density is the first-order constraint on whether a body is made of rock, ice, or gas. Used throughout [Lecture 8](#) and [Lecture 13](#).

6. Hydrostatic equilibrium

$$\frac{dP}{dr} = -\rho g$$

The pressure gradient that balances gravity in any static fluid layer. It structures both atmospheres and interiors and is the starting point of the scale-height and central-pressure estimates. Introduced in [Lecture 5](#) and [Lecture 8](#).

7. Ideal gas law

$$P = nk_B T = \frac{\rho k_B T}{\mu m_u}$$

The equation of state of a dilute gas, with number density n , mean molecular weight μ , and atomic mass unit m_u . It closes the hydrostatic equation for atmospheres. Introduced in [Lecture 5](#).

8. Isothermal scale height

$$H = \frac{k_B T}{\mu m_u g}$$

The e-folding height of pressure in an isothermal atmosphere, from hydrostatic equilibrium plus the ideal gas law. Hotter or lighter atmospheres are more extended; stronger gravity compresses them. Derived in [Lecture 5](#).

9. Stefan–Boltzmann law

$$F = \sigma T^4$$

The radiative flux from an ideal (blackbody) surface at temperature T , with σ the Stefan–Boltzmann constant. [Lecture 3](#) states the general form $F = \epsilon \sigma T^4$, with emissivity $\epsilon \leq 1$ close to 1 for most planetary surfaces. It governs how planets shed heat to space.

10. Equilibrium temperature (energy balance)

$$\frac{S}{4}(1 - A) = \sigma T_{\text{eq}}^4 \quad \Rightarrow \quad T_{\text{eq}} = \left[\frac{S(1 - A)}{4\sigma} \right]^{1/4}$$

Absorbed stellar flux (stellar constant S , Bond albedo A , the factor 4 from the ratio of a sphere's surface to its cross-section) equals emitted thermal flux. This balance sets the baseline temperature of every planet before greenhouse warming. Derived in [Lecture 5](#); extended to the greenhouse model in [Lecture 9](#) and applied to exoplanets in [Lecture 13](#).

11. Thermal energy per particle

$$E_{\text{th}} = \frac{3}{2} k_B T$$

The mean kinetic energy of a gas particle at temperature T . Comparing thermal and gravitational energy per molecule decides between retention and escape of an atmosphere. The escape physics of [Lecture 5](#) and [Lecture 10](#) builds on this comparison; the Jeans parameter there uses $k_B T$ as the thermal scale.

12. Radioactive decay

$$N(t) = N_0 e^{-\lambda t}, \quad t_{1/2} = \frac{\ln 2}{\lambda}$$

The exponential decay of a parent isotope with decay constant λ and half-life $t_{1/2}$. It powers radiogenic heating of interiors and underlies all radiometric dating. The half-lives behind radiogenic heating appear in [Lecture 3](#); the decay law itself is introduced with radiometric dating in [Lecture 12](#).